QUANTITATIVE PLANETARY IMAGE ANALYSIS / RESEARCH ARTICLE

Subpixel Geometric Reconstruction of a Cylindrical Candidate in Curiosity Mastcam-100 Imagery

Boundary metrology, frozen-shape prediction, and geological comparison

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Curiosity Sol 3556 / `3556MR1025170721700585C00`

Abstract

A centimeter-scale object in Curiosity Mastcam-100 imagery acquired on sol 3556 exhibits two curved terminations and a straight upper exterior boundary. Physically directed subpixel localization yields 35 near-rim, 24 far-termination, and 26 upper-boundary stations. A six-parameter common-section, common-axis model with exact upper tangency fits all 85 observations at 0.3788 px RMS. More importantly, a front/dorsal-derived ellipse with frozen size, shape, and orientation predicts rear points beyond those used for translation: six anchors predict the other 18 stations at 0.2084 px RMS, and guarded contiguous rear holdouts yield 0.2429 px. Guarded spatial holdouts of the strict model yield 0.7233 px across all three features. Withholding the entire dorsal boundary is reported separately as a cross-feature sensitivity test. A completed comparison of 28 Curiosity images, including 18 uniformly measured whole frames, demonstrates comparable straight segments in geological scenes but does not establish the frequency of the full measured conjunction. The configuration exhibits manufacturing-like geometric regularity supported by predictive tests. A finite-boundary sensitivity also admits a physically tangent tapered segment, leaving the complete three-dimensional shape and origin unresolved. The archived-image workflow provides a reproducible benchmark for comparative morphology and physical-model testing.

Keywords: Mars; Gale crater; Curiosity; Mastcam-100; cylinder; subpixel metrology; orthogonal regression; image geometry; occlusion; concretion; technosignature assessment.

1. Introduction

A photograph can establish the projected geometry of an object even when its composition, depth, and origin remain unknown. For the Curiosity Mastcam-100 observation considered here, the central question is unusually specific: can the distinct curved boundary at the foreground end, the exposed curvature at the far end, and the upper longitudinal boundary be represented by one common cylindrical geometry? The central evidentiary question is whether that geometry predicts observed boundary points that did not participate in its registration or fit. Determining physical origin is beyond the present scope.

The target occurs in the image identified by `3556MR1025170721700585C00`, acquired on sol 3556 (7 August 2022). The Mastcam-100 instrument is a fixed-focal-length, narrow-field camera with an instantaneous field of view near $74\,\mu$rad per image pixel, making subpixel analysis appropriate for centimeter-scale objects at several meters' range [3, 4]. The target is not a full, free-standing object on a uniform background: its lower outline merges with the regolith, and a dark region adjacent to the near end contains a cast-shadow contribution. Consequently, not every visible intensity transition belongs to the solid's perimeter.

We define material boundaries independently of their suitability for any candidate shape. The upper edge is selected at the bright-body/dark-regolith interface; the lower visible contact is classified as possible foreground soil occlusion rather than automatically being treated as a second silhouette generator. Both terminal curves are registered with explicit parameter constraints. The measured feature families are then tested jointly under common-section and tangency constraints. The image data and executable analyses are retained in the research repository [1].

This paper reports measurements, predictive validation, uncertainty structure, and the completed geological comparison. To our knowledge, this is the first publicly documented, reproducible quantitative metrology study of this particular feature. Earlier recognition and visualization are acknowledged: Ward credits Rami Bar Ilan with discovery, and MILTOID presented Bayer decoding and a visual 3D cylinder overlay in October 2025 [10, 11]. Those preliminary checks precede the numerical metrology and predictive tests reported here. The original study appears in the repository history on 24 October 2026 and in the Zenodo deposit published 30 October; independent reconstruction was added on 11 November, followed by the 13 November predictive validation [12].

2. Observational Material and Provenance

2.1 Instrument products and pixel conventions

The primary inputs are NASA PDS3 Reduced Data Record products from the right Mastcam, with product stem `3556MR1025170721700585C00` and dataset designation `MSL-M-MASTCAM-4-RDR-IMG-V1.0`. The investigation operates on zero-based image coordinates (x, y), with x increasing rightward and y increasing downward. For the principal radiometric product, an image coordinate refers to the array location in the PDS subframe, not the full detector's absolute position.

Table 1: Image products and their analytic roles. Dimensions are ordered as image lines × samples × bands.

Suffix	Array shape	Encoding	Use in analysis
`DRXX`	$1184 \times 1328 \times 3$	big-endian 16-bit	Primary radiometric pixel measurements; native CAHVOR geometry
`DRLX`	$1180 \times 1323 \times 3$	big-endian 16-bit	Independently sampled linearized CAHV geometry
`DRCX`	$1184 \times 1328 \times 3$	8-bit	Tonal-display and localization reference
`DRCL`	$1180 \times 1323 \times 3$	8-bit	Linearized display and visual comparison

The image metadata give an acquisition time of 2022-08-07 20:58:23.349 UTC, local true solar time 12:10:25, a 14.2 ms exposure, and clear-filter Mastcam-right acquisition. The principal data were transmitted with predictive lossless Bayer Huffman compression. Their geometric features therefore are not attributable to JPEG block boundaries. Demosaicing, optical blur, radiometric scaling, and companding nevertheless influence local edge localization and are addressed through multi-band and estimator-sensitivity comparisons.

The per-band radiometric conversion implemented in the repository is

$$I_b(x, y) = o_b + q_b \, \mathrm{DN}_b(x, y), \quad (q_R, q_G, q_B) = (1.201, 1.112, 1.037) \times 10^{-4}, \tag{1}$$

with $o_b = 0$ in the processed products and radiance in W m^{-2} sr^{-1} nm^{-1}. The mean of the three band radiances defines the primary scalar field $I(x, y)$; separate-channel fits provide a check on color interpolation effects.

2.2 Four distinguishable visual regions

The segmentation distinguishes (i) the sunlit longitudinal surface, (ii) its upper background-facing edge, (iii) the foreground curved termination separating illuminated material from the dark recess, and (iv) the lower-right dark patch, whose distal boundary partly follows cast shadow on soil rather than the object's cross-section. The far upper-left termination has its own observable material/background boundary, while its lower part is obscured by regolith.

This classification is causally important. An elliptical fit to a shadow boundary measures illumination geometry and ground topography, not necessarily the perimeter of the object that cast the shadow. Likewise, a line running through the brightest surface is not necessarily the outer silhouette, even if its projected slope is nearly constant.

The analysis reports a regolith-noise estimate $\sigma_I = 7.782 \times 10^{-3}$ in its radiometric units, a sunlit-object mean near 9.848×10^{-2}, and a dark-core mean near 1.358×10^{-2}, with an edge signal-to-noise estimate of 10.91 for its selected contrast definition. These measurements characterize signal separation without assuming anything about the object's material.

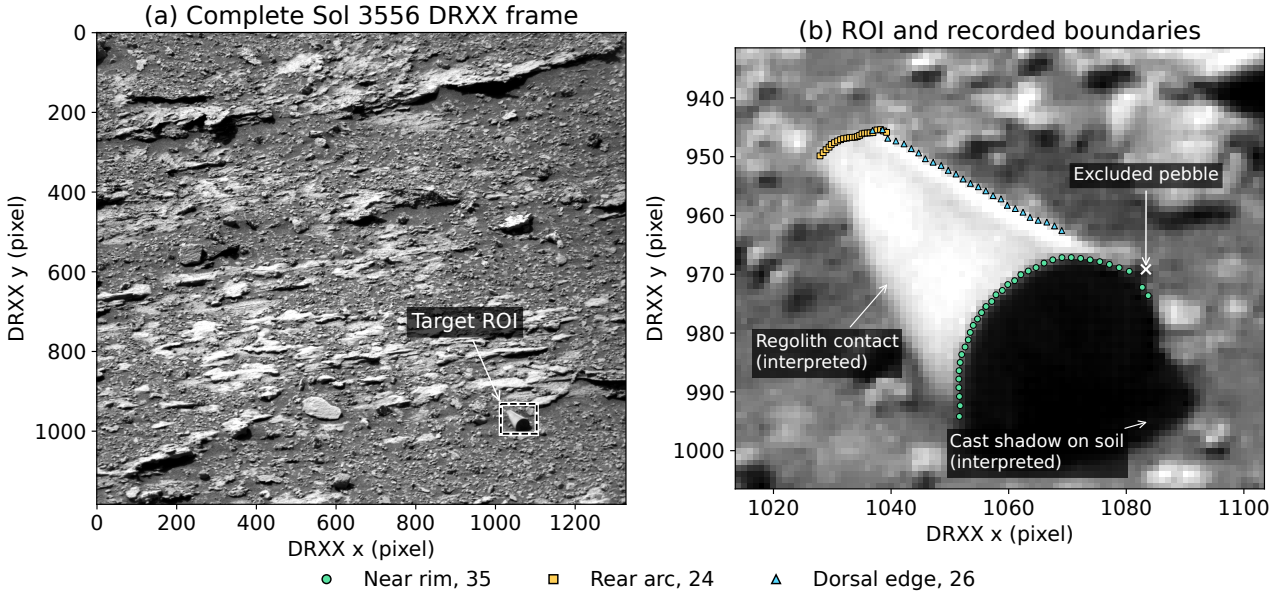


Figure 1: Authentic Sol 3556 Mastcam-100 DRXX radiance. (a) Complete 1328×1184-pixel PDS frame with the target ROI marked at lower right. (b) Enlargement of the half-open crop $x \in [1014, 1104)$, $y \in [932, 1007)$, retaining the recorded boundary stations and excluded pebble. Axes use zero-based DRXX subframe coordinates. The frame and crop use separate linear grayscale limits (1st–99th and 2nd–98th percentiles, respectively), without spatial interpolation. Soil and shadow labels identify interpreted regions. Product `3556MR1025170721700585C00_DRXX`; image credit NASA/JPL-Caltech/MSSS.

3. Camera Geometry, Illumination, and Scale

3.1 CAHV and CAHVOR projection

Let **C** be the camera center, **A** its optical axis, and **H**, **V** the horizontal and vertical projection vectors. In the linear CAHV model, a three-dimensional point **P** projects to

$$x = \frac{(\mathbf{P}-\mathbf{C})\cdot\mathbf{H}}{(\mathbf{P}-\mathbf{C})\cdot\mathbf{A}}, \qquad y = \frac{(\mathbf{P}-\mathbf{C})\cdot\mathbf{V}}{(\mathbf{P}-\mathbf{C})\cdot\mathbf{A}}. \tag{2}$$

The corresponding nonlinear CAHVOR model applies radial corrections about distortion axis **O**:

$$\mathbf{p} = \mathbf{P}-\mathbf{C}, \qquad \omega = \mathbf{p}\cdot\mathbf{O}, \tag{3}$$

$$\lambda = \frac{\mathbf{p}}{\omega} - \mathbf{O}, \qquad \tau = \lambda\cdot\lambda, \tag{4}$$

$$\mu = R_0 + R_1\tau + R_2\tau^2, \qquad \mathbf{p}' = \mathbf{p} + \mu(\mathbf{p} - \omega\mathbf{O}). \tag{5}$$

The corrected vector $\mathbf{p}'$ replaces **p** in eq. (2). The script also inverts image coordinates to viewing rays and transforms rover-navigation vectors to the local site reference frame. The independently derived linearized camera azimuth and elevation agree with the recorded pointing values of 227.7412° and -22.0047°, respectively.

At the target location near $(1061, 980)$, the singular values of the local angular-to-image Jacobian are $13{,}498.24$ and $13{,}418.43\,\mathrm{px\,rad^{-1}}$. Accordingly,

$$f_{\mathrm{local}} = 13{,}458.28\,\mathrm{px\,rad^{-1}}, \quad \mathrm{IFOV} = 74.304\,\mu\mathrm{rad\,px^{-1}}, \quad \epsilon_{\mathrm{anisotropy}} = 0.595\%. \tag{6}$$

The small local anisotropy means that a major departure from circularity cannot reasonably be attributed to the lens at this image location.

3.2 Range-sensitive physical dimensions

Without an overlapping target stereo solution, angular and pixel-based measurements are better constrained than dimensional measurements in millimeters. Using the metadata's best focus distance $D_f = 3.957\,\mathrm{m}$ as a working range gives a local sampling scale

$$\ell_{\mathrm{px}}(D) \approx D\,(74.304\times10^{-6}) = 0.2940\left(\frac{D}{3.957\,\mathrm{m}}\right)\mathrm{mm\,px^{-1}}. \tag{7}$$

Within the recorded focus search bracket 3.6–4.4 m, this scale spans approximately 0.2675–$0.3269\,\mathrm{mm\,px^{-1}}$. A projected strict-model diameter of $2a \approx 39.99\,\mathrm{px}$ therefore corresponds to 10.70–13.07 mm across that bracket, or $11.76\,\mathrm{mm}$ at the focus-distance scenario. The projected center separation of $40.86\,\mathrm{px}$ corresponds to $12.0\,\mathrm{mm}$ at that same assumed distance. These illustrative physical conversions scale linearly with assumed range; they are not stereo-derived dimensions. Projected center separation is distinct from the full three-dimensional object length, which requires an additional orientation estimate.

3.3 Solar geometry and cast-shadow direction

PDS metadata give a local solar azimuth of 186.8169° and elevation of 69.8224°. Projection through the camera model gives a solar image-plane direction near -116.571° and a corresponding shadow direction near $+63.429^\circ$. This direction is consistent with the extensive lower-right dark region. Modeling the shadow as an optical projection onto an underlying soil surface is essential for avoiding an incorrect full-outline fit: the shadow's distal edge need not be concentric with the exposed curved material boundary.

In a simple receiving-plane formulation, let the shadow-casting rim point be $\mathbf{P}$, the unit Sun-direction vector be $\hat{\mathbf{s}}$, and the soil plane satisfy $\hat{\mathbf{n}}_g \cdot (\mathbf{X} - \mathbf{P}_0) = 0$. The physical shadow intersection is

$$\mathbf{X}_{\text{shadow}} = \mathbf{P} - \lambda \hat{\mathbf{s}}, \qquad \lambda = \frac{\hat{\mathbf{n}}_g \cdot (\mathbf{P} - \mathbf{P}_0)}{\hat{\mathbf{n}}_g \cdot \hat{\mathbf{s}}}, \tag{8}$$

when $\lambda > 0$. Camera projection of the resulting point gives the expected image location of the cast shadow. The receiving-plane orientation is a modeled scene property, not an independently measured surface normal. The construction nevertheless explains why the terminal material curve and the lower distal shadow boundary need not be pieces of the same conic.

4. Subpixel Measurement of Physical Boundaries

4.1 Normal-profile extraction

For a line segment with unit tangent $\hat{\mathbf{t}} = (t_x, t_y)$, define its directed normal $\hat{\mathbf{n}} = (-t_y, t_x)$. Intensity is sampled on rays

$$\mathbf{x}_i(s) = \mathbf{x}_{i,0} + s\hat{\mathbf{n}}, \qquad g_i(s) = I\big(\mathbf{x}_i(s)\big), \tag{9}$$

with bilinear interpolation on the radiometric image. After mild Gaussian smoothing, the derivative $g_i'(s)$ is searched for a contrast extremum of the *physically expected sign*. A local three-sample parabola refines the position; the surrounding unsmoothed profile is also fitted to an error-function transition,

$$g(s) = I_- + \frac{I_+ - I_-}{2}\left[1 + \operatorname{erf}\left(\frac{s - s_0}{\sqrt{2}\,\sigma_{\text{PSF}}}\right)\right], \tag{10}$$

where s_0 is the transition coordinate, σ_{PSF} a profile-width parameter, and $I_\pm$ the fitted asymptotic intensities.

For the upper edge, the selected normal runs from darker background into brighter illuminated object. Its appropriate gradient search polarity is $+1$. For the lower edge it runs from brighter object into darker soil, requiring -1. The implementation records the sign-adjusted intensity difference across each selected station and checks that the transition follows the expected material boundary.

4.2 Exterior-boundary validation

The upper boundary is defined by the large, spatially coherent radiometric transition from background regolith into the illuminated body. At each of 26 stations, the extractor records transition contrast, gradient direction, fitted transition position, and fit uncertainty. All 26 stations satisfy the expected polarity diagnostic. Their best-fit line is at 28.895°, with 0.2422 px orthogonal RMS over approximately 36.4 px of the visible exterior. The edge-coordinate arrays and diagnostic fields are retained in `step3_features.json`. This feature definition is independent of the subsequent geometric fits.

4.3 Near rim: separation from the adjacent pebble

The near curved termination is sampled at five-degree radial intervals. Of 36 near-rim stations, precisely one at $\theta = -45^\circ$ lies across a narrow shadow gap on the adjacent upper-right pebble. Its extracted position is approximately $(1083.34, 969.16)$, radius 22.40 px, compared with neighboring radii of 19.86 and 20.34 px. The near-rim dataset therefore comprises the 35 stations that lie on the continuous observed termination, with the neighboring pebble excluded by spatial and contrast classification.

4.4 Far termination: separating cap, soil, and longitudinal transition

The rear region is sampled from an interior seed at $(1035.5, 951.2)$, with radial angles spanning -210° to -35°. The extractor selects the first outward crossing of a radiometric threshold $I_{\text{iso}} = 0.0860$ before fitting the local transition. The crucial physical partition is between the buried lower-left contact (indices 0–7), the exposed rear cap (indices 8–31), and the upper longitudinal transition (indices 32–35). Only the 24 middle samples are treated as the terminating curve. This prevents a soil-occlusion line from masquerading as a noncircular end.

5. Curve Fitting and Geometric Models

Figure 2 registers the retained stations with the strict model on the original radiometric crop. Dashed section extensions distinguish the fitted complete curves from their sampled intervals.

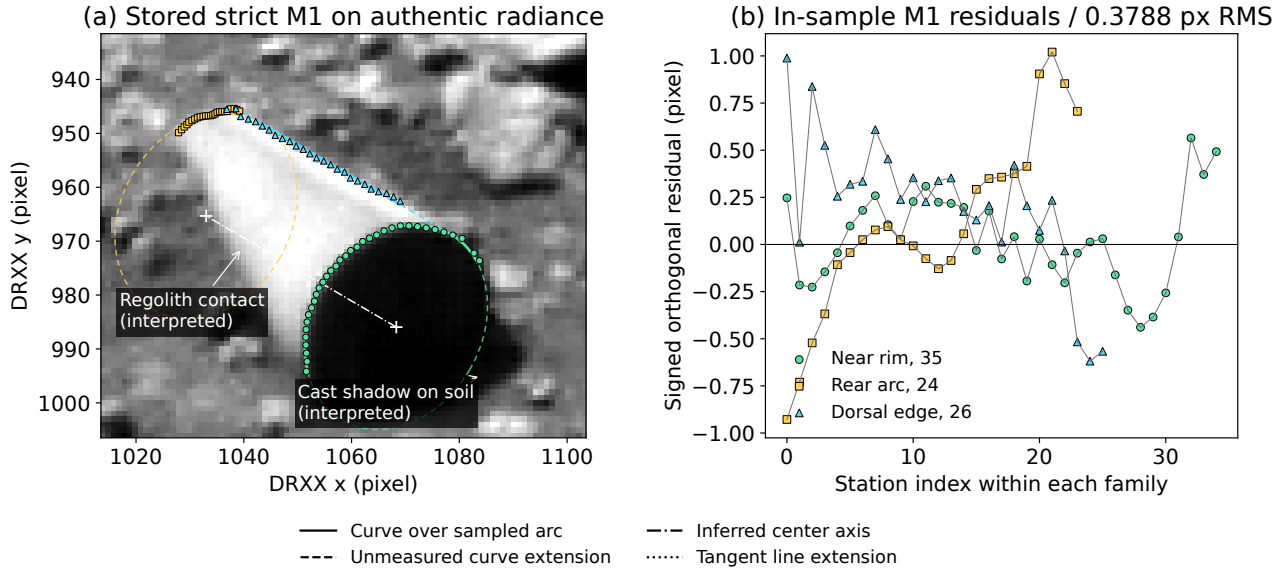


Figure 2: Stored six-parameter M_1 geometry and in-sample residuals. (a) The original DRXX crop carries all 35 near-rim circles, 24 rear-arc squares and 26 dorsal triangles. Solid section curves span the sampled arcs; dashed curves continue unmeasured geometry. Crosses and the dash-dot line mark inferred centers and their common axis. The upper tangent joins the modeled terminal tangencies; its dotted continuation shows the line extension used by the fitting objective. Regolith-contact and cast-shadow labels are interpretations. (b) Signed orthogonal residuals from the exact parameters in `step9_joint_model.json`, with joint RMS 0.3788 px. Stations and fitted curves share the acquisition and are distinct from the withheld predictions in fig. 3. Display uses the same crop transform as fig. 1; image credit NASA/JPL-Caltech/MSSS.

5.1 Orthogonal-distance estimation

Straightness is estimated using total least squares. For observations $\{\mathbf{x}_i\}$, the eigenvector of the smaller eigenvalue of the 2D sample covariance gives the line normal $\hat{\mathbf{n}}$. Its signed orthogonal residual is

$$e_i = \hat{\mathbf{n}} \cdot (\mathbf{x}_i - \bar{\mathbf{x}}), \qquad \mathrm{RMS}_\perp = \sqrt{N^{-1} \sum_i e_i^2}. \tag{11}$$

For the end curves, an image-plane ellipse centered on $\mathbf{c}$ is parameterized in coordinates aligned with a trial body axis θ:

$$\hat{\mathbf{u}} = (\cos\theta, \sin\theta), \qquad \hat{\mathbf{v}} = (-\sin\theta, \cos\theta), \tag{12}$$

$$\mathbf{x}(\psi) = \mathbf{c} + b\cos\psi\,\hat{\mathbf{u}} + a\sin\psi\,\hat{\mathbf{v}}, \tag{13}$$

where a is the transverse semi-major axis and b the longitudinally aligned semi-minor axis. A point's residual is its minimum Euclidean distance to the curve, rather than a raw algebraic-conic residual. Newton refinement provides the closest curve point within each nonlinear least-squares evaluation.

5.2 Uncoupled reference fits

The 35 clean near-rim observations produce a freely oriented ellipse with $a = 19.7170$ px, $b = 16.4240$ px, $b/a = 0.8330$, and minor-axis angle 32.704°, with 0.16395 px RMS. Holding that shape and orientation fixed while translating only its center fits the 24-point rear arc at 0.18629 px RMS. The translated rear center is approximately $(1035.98, 964.48)$, relative to the near center $(1068.88, 986.00)$, giving a center-to-center bearing of 33.185° and projected separation of 39.32 px. This independent two-parameter registration is important because the rear shape is not freely redrawn.

5.3 Six-parameter strict-cylinder construction

The strict model M_1 ties all three observed families to a common, projected constant-section body. Its parameter vector is

$$\mathbf{p}_1 = (x_f, y_f, a, b, L, \theta), \tag{14}$$

with equal near/far ellipse parameters and collinear centers

$$\mathbf{c}_r = \mathbf{c}_f - L\hat{\mathbf{u}}(\theta), \qquad (a_r, b_r) = (a, b). \tag{15}$$

The upper image boundary is further constrained to be the outer tangent of that same section:

$$\hat{\mathbf{v}}(\theta) \cdot (\mathbf{x}_d - \mathbf{c}_f) = -a. \tag{16}$$

Thus there is *no independently adjustable upper-line offset*. Let F index the near rim, R the far arc, and D the upper boundary. The joint least-squares objective is

$$\chi^2(\mathbf{p}_1) = \sum_{i\in F} \left(\frac{e_{F,i}}{\sigma_{F,i}}\right)^2 + \sum_{i\in R} \left(\frac{e_{R,i}}{\sigma_{R,i}}\right)^2 + \sum_{i\in D} \left(\frac{e_{D,i}}{\sigma_{D,i}}\right)^2. \tag{17}$$

Raw pixel RMS is reported separately as

$$\mathrm{RMS}_{\mathrm{joint}} = \sqrt{\frac{\sum_{i \in F \cup R \cup D} e_i^2}{85}}. \tag{18}$$

This ensures that differences in the formal uncertainties σ_i do not get silently confused with the reported geometric residuals.

The resulting fit parameters are

$$\begin{aligned} (x_f, y_f) &= (1068.3014, 985.9459), \qquad (x_r, y_r) = (1033.0267, 965.3205), \\ a &= 19.9925\,\mathrm{px}, \quad b = 15.7391\,\mathrm{px}, \quad L = 40.8622\,\mathrm{px}, \\ \theta &= 30.3152^\circ, \qquad s = a_r/a_f = 1 \quad (\text{fixed}). \end{aligned} \tag{19}$$

The fitted projected geometry is highly constrained: a common orientation, equal section size and shape, collinear centers, and an exact upper tangent are enforced simultaneously. In image-space terms, these are precisely the relations expected of a projected cylindrical body. The observational model is a coupled two-dimensional geometric representation, not an independently triangulated 3D surface scan.

5.4 More flexible comparison models

Two constant-section variants clarify which constraints are doing work. M_2 retains equal cross-sections and collinearity but estimates a separate upper-crest offset r_d, increasing parameter count from six to seven. M_3 searches projected-axis angle in 0.1° increments, minimizing unweighted residuals while also allowing r_d; because the angle is selected from the data, it has seven effective fitted quantities. The earlier tapered formulation M_4 is retained only as a rejected construction in section 6.6.

6. Results

6.1 The joint result across all 85 measured points

Table 2 summarizes the three constant-section joint fits. The unweighted raw RMS of strict M_1 is 0.3788 px, partitioned into near-rim 0.2376 px, far-arc 0.4830 px, and upper-edge 0.4217 px. Models with an adjustable upper offset yield 0.3468 px (weighted M_2) or 0.3389 px (unweighted M_3).

Table 2: Current joint-fit results from `step9_joint_model.json`; $N = 35 + 24 + 26 = 85$ observations. RMS figures are unweighted orthogonal residuals in native image pixels. Model M_3 counts its scanned angle as a fitted parameter.

Fit	k	s	θ	Near	Far	All
M_1, strict tangent	6	1.000	30.32°	0.238	0.483	**0.379**
M_2, offset	7	1.000	31.10°	0.231	0.305	0.347
M_3, unweighted offset	7	1.000	30.60°	0.242	0.378	0.339

The corresponding upper-edge RMS values are 0.422, 0.485, and 0.405 px, respectively. For M_1, $\chi^2 = 2653.51$ with 79 nominal residual degrees of freedom gives $\chi^2_\nu = 33.59$, indicating that the formal station errors do not supply a calibrated likelihood.

The six-parameter model's residuals reflect its simultaneous geometric obligations. It simultaneously puts the rear and near curves on one axis, fixes their common section size, and makes the measured upper exterior edge tangent to that section. Its residuals therefore constitute positive evidence for consistency with the specified

common-section image-plane geometry. The finite extent of its tangent boundary is evaluated separately below.

6.2 Near- and far-termination agreement

Independently of the joint fit, the translated unchanged near ellipse matches the far arc at 0.18629 px RMS. The far curve covers only part of the termination; it has no requirement to expose its complete lower perimeter. This registration is therefore evidence that the visible part of the far terminus can carry the same local curvature as the near rim, not an observation of two complete closed end faces.

The test is stronger than a statement that each curved boundary can be approximated by an unrelated ellipse: shape, orientation, and scale are held common. It is weaker than a stereo reconstruction of both terminal planes. The distinction follows directly from the degrees of freedom and the observation geometry.

6.3 Predictive validation beyond registration points

The new tests preserve the existing coordinates and exclusions. Their protocol was written before these validation runs; the tests are retrospective coordinate-level checks, not a blind repeat of boundary selection. Two front-shape references must be distinguished. The freely oriented ellipse above gives 0.18629 px rear RMS. The predictive tests instead freeze the front ellipse constrained to the measured dorsal orientation: $a = 20.02897$ px, $b = 16.84577$ px, $\theta = 28.89475^\circ$, and scale $s = 1$. Fitting only its rear translation reproduces 0.1854 px RMS. Neither rear size nor orientation is adjusted.

Table 3: Descriptive fits and heldout predictions. All errors are raw RMS distances in native pixels; the first and fourth rows are in-sample references.

Check	Used in fit	Scored observations	RMS
Frozen rear registration	All 24 rear stations; two translations	Same 24 rear stations	0.1854
Sparse registration	Rear indices 0, 4, 8, 12, 16, 20	Other 18 rear stations	0.2084
Guarded rear blocks	Remaining rear stations, excluding guards	Each rear station withheld once	0.2429
Strict M_1, full fit	All 85 stations; six parameters	Same 85 stations	0.3788
Guarded M_1 blocks	Remaining stations from all three families	Each of 85 stations withheld once	0.7233
Entire dorsal withheld	All front and rear stations only	All 26 dorsal stations	1.9092

In five contiguous rear folds, the test block and two neighboring guard stations at each boundary are excluded from translation fitting. The strict-model folds similarly withhold one contiguous block from each feature family, with two-station guards. Every point is scored once across folds. Three prespecified starting angles (28°, 31.2°, 35°) are tried; the smallest training objective selects the solution, never the withheld score. All starts converge and no parameter reaches a bound. Strict-model heldout RMS values are front 0.8456 px, rear 0.5946 px, and dorsal 0.6491 px; pooling their squared residuals gives 0.7233 px.

The rear agreement survives both sparse registration and guarded holdouts at approximately 0.2–0.24 px. It therefore extends beyond the rear points used to position the transferred shape. This predicts local curve positions after registration, not the rear's absolute position or cylinder length from the front alone.

Withholding the entire dorsal feature is a separate sensitivity experiment. Front/rear-only fitting predicts its tangent at 1.9092 px RMS, maximum 3.391 px, with axis 33.177° rather than the measured dorsal bearing 28.895°. These modest absolute image deviations are larger than the residuals of the complete fit. They demonstrate the information supplied by the dorsal boundary: removing that measured constraint changes the preferred

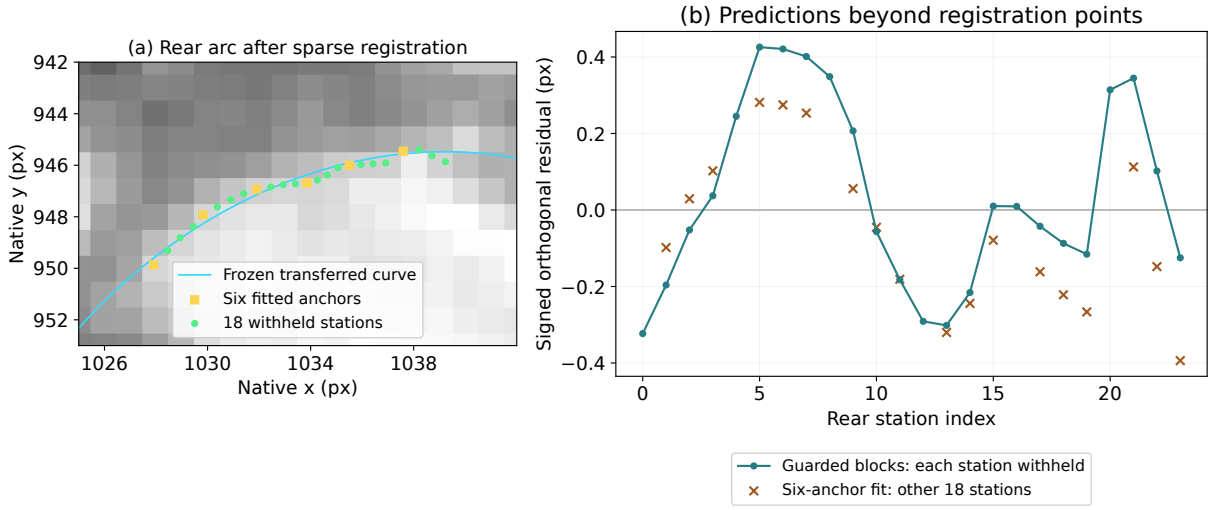


Figure 3: Frozen front/dorsal-derived ellipse translated using six rear anchors. Left: native radiometric crop with the transferred curve, fitted anchors, and 18 withheld stations. Right: signed orthogonal residuals for sparse-registration and guarded-block predictions. The curve is clipped to the exposed rear region for legibility; the concealed perimeter is not observed.

orientation and tangent. This experiment is not a mandatory passing standard, a measurement of physical surface damage, or a calibrated rejection. It also does not replace the fixed-shape rear tests, which retain the dorsal information by design.

6.4 Axis alignment and orientation identifiability

The separate estimates for the near-ellipse minor axis, far-to-near center bearing, and upper line are 32.704°, 33.185°, and 28.895°, respectively. The near ellipse is observed along an incomplete curve, allowing its angle to vary under small changes of the selected points. The reported bootstrap spread for the near-ellipse orientation is approximately 4.23°. A transverse shift of $1\,\text{px}$ over a $40.6\,\text{px}$ center baseline changes its bearing by approximately 1.41°. These sensitivities put the several-degree differences in context. Joint optimization chooses 30.315° for the strict cylinder, and achieves subpixel residuals on each point family.

6.5 Occlusion versus intrinsic taper

The upper exterior boundary and lower visible diagonal need not represent two opposite silhouette generators. Their measured apparent angular convergence is approximately 35°, whereas perspective convergence between parallel generators at the measured camera scale is below 0.4° for the considered axis inclinations. This excludes the interpretation of *both visible lines* as the unoccluded opposite sides of a parallel-sided body, but does not exclude a partially buried cylinder.

In the partially buried geometry, the upper transition is a material silhouette, while the lower diagonal is the contact between the body and foreground regolith. The latter can cut obliquely across a cylindrical surface as burial depth changes along its length. The two image lines therefore encode different physical boundaries, and their angular separation is not an estimator of cylinder taper without first establishing that both are true generators.

6.6 Rear-scale sensitivity and rejected tapered construction

A *standalone* fit translates an otherwise fixed near-ellipse shape and optionally scales it by s, using only 24 exposed rear points. Its minimum occurs at $s \approx 0.93$ with 0.18434 px RMS, while the exact equal-size case $s = 1$ yields 0.18629 px, a difference of 0.00195 px. Across $s = 0.85$–1.15, the variations in raw residual remain small in the context of the image sampling. The conditional profile-likelihood interval recorded by the pipeline is $[0.87, 0.99]$; it does not literally include $s = 1$ under that specific weighting model. Raw-fit proximity and formal confidence limits are different statements.

The earlier eight-parameter formulation M_4 is rejected as a physical frustum despite its 0.2549 px numerical RMS. Its proposed boundary $v = -r - mu$, where $m = (1-s)r/L$, cuts its own end ellipse: tangency requires $r^2 = a^2 + m^2b^2$, but its fitted intercept is 19.294848 px rather than the required 19.433711 px. Its parameters and residuals remain in `step9_joint_model.json` for traceability; they are not treated as a viable physical alternative. This failure concerns that particular formulation, not every possible tapered body.

6.7 Finite-boundary and camera-projected shape sensitivity

The line residual in eq. (16) allows an indefinitely extended tangent. Five dorsal stations project beyond the rear endpoint of the stored M_1 generator, as the dotted extension in fig. 2 shows. The original joint-fit and holdout results quantify that stated line-and-conic model. They do not impose a finite-generator endpoint constraint.

A subsequent exploratory check projects finite circular and tapered segments through the recorded CAHVOR camera. Range is fixed at 3.957 m. Both models fit the center ray, radius, axial length and orientation; the frustum additionally fits its rear/front radius ratio. The side boundary follows surface tangency, with no independent upper offset. Refitting with Euclidean distance to the finite generator, using the retained stations, weights, starting bearings and guarded folds, gives table 4. These are separate physical-model sensitivities. The original M_1 results and the frozen-shape rear tests retain their definitions.

Table 4: Exploratory CAHVOR-projected models with finite-generator distances. Full fit uses all 85 stations; guarded prediction withholds each station once. Scores are native-pixel RMS.

Surface	k	Full fit	Guarded
Circular segment	6	0.7717	1.9363
Tapered segment	7	0.3946	0.7176

The finite frustum fits a radius ratio of 0.4548. Its geometry satisfies the tangency condition that the earlier M_4 construction violated. The fixed-front rear-scale scan above cannot supply a rejection significance for this jointly changed front shape, orientation and rear center. Its lower coordinate errors admit taper as an exploratory shape alternative; complete terrain occlusion, cast shadows and radiance have not been jointly fitted. A three-parameter soil plane fitted to the circular segment's contact coordinates illustrates the distinction: its contact-curve RMS is 0.1822 px, yet one rendered configuration hides the observed near rim and dorsal boundary. A subsequent finite foreground-bank illustration assumes terrain depth. Detailed endpoint, visibility and alternative-model records are retained under `audit/sediment-pipe-comparison/`.

7. Sensitivity and Repeatability

Measurement stability is assessed not merely by the visual appearance of fitted curves but by changing the estimation rule while keeping the scene fixed. The compiled output compares error-function transition centers, parabolically interpolated gradient maxima, and integer-pixel stations. The two subpixel methods return nearly the same upper-edge angle and consistently smaller line residuals than the discrete alternative (table 5). An additional trimming experiment removes stations near the segment endpoints; the upper-line orientation shifts moderately while its central portion becomes straighter, as expected if boundary transitions are most ambiguous near neighboring objects and termini.

Table 5: Sensitivity measurements in the committed `measurements.json`. Values reflect alternative estimators and restricted subsets of the same target, not independent Mars images.

Measurement procedure	θ_{upper}	Upper-line RMS	Interpretation
Error-function subpixel	28.895°	$0.2422\,\text{px}$	Selected exterior boundary
Parabolic gradient peak	28.933°	$0.2105\,\text{px}$	Close independent localization rule
Integer-pixel positions	29.105°	$0.4559\,\text{px}$	Quantization inflates residual
Trim outer 15%	29.154°	$0.1742\,\text{px}$	Central line remains straight
Trim outer 25%	29.609°	$0.1195\,\text{px}$	Fewer stations and shorter baseline

In separate synthetic correlated-noise perturbations, the reported standard deviation of the apparent lower/upper convergence grows from 1.08° at one times empirical noise to 2.45° at three times noise. The method is therefore sensitive to optical sampling and pixel covariance, but the resulting uncertainty can be investigated explicitly rather than hidden behind the assumption of perfect edge localization. These tests characterize the stability of selected image boundaries. Physical tolerance and error-model limits are collected in section 9.2.

8. Texture, Straightness, and Illumination as Independent Constraints

The selected exterior edge has $0.2422\,\text{px}$ orthogonal line RMS over a sampled span of approximately $36.4\,\text{px}$. At the focus-distance reference scale that corresponds to an image-plane residual length of about $71\,\mu\text{m}$. This is *a projected measurement residual*, not profilometric roughness of a three-dimensional surface at micrometer resolution. A quadratic curvature estimate for the same edge gives a maximum sagitta of $0.280\,\text{px}$.

The completed comparison uses 28 retrieved Curiosity images, including 18 whole Mastcam frames measured uniformly and ten captioned geological/context images. A common contour method applies 2×2 area averaging, brightness-level contours, and 26 equally spaced samples over a 36.4-native-pixel span. Its target window is selected by proximity to the existing dorsal stations, not minimum RMS. It yields $0.303\,\text{px}$ RMS, a separate measurement from the full-resolution erf result of $0.2422\,\text{px}$.

Among 226,426 control windows, 835 (0.369%) are at least as straight; qualifying segments occur in 17 of 18 frames. Straight segments therefore occur in geological scenes. These overlapping photometric windows are correlated and may follow material, fracture, shadow, or shading boundaries; they are not counts of independent rocks. Their fraction is not an origin probability. Smoothing and span change the conditional fraction, so the full sensitivity tables are retained in the supplementary comparison report [13].

A separate original-image screen retrieved nine PDS views covering the four Gale crater pipe localities discussed by Rubin et al. [7, 8]. Three partial-rim views of two objects received two-method boundary extraction. No screened view exposes both axial terminations and a connecting exterior boundary of one object for a

matched joint test. Local texture and fragmented contacts also make some extracted stations ambiguous. These partial measurements provide an observational inventory; they do not quantify the frequency of the target's complete geometric configuration or reject the sediment-pipe mechanism as a class.

The conjunction of terminal curvature, transferred geometry, and longitudinal constraint remains the relevant object-level result. The comparison did not estimate the frequency of that complete conjunction. Its automatic rim contour differs from the recorded rim stations by 2.88 pixels on average, preventing a comparable rim ranking; appearance-search scores likewise are not physical-model likelihoods. No full counterpart of the target's 85-station reconstruction was demonstrated by that study. The detailed image inventory, comparison atlas, preprocessing sensitivity, and appearance retrievals remain supplementary material.

Three locally selected natural rock boundaries in the same image provide secondary context. Their reported line RMS values are 0.971, 1.724, and 1.953 px, respectively, compared with 0.242 px for the target (table 6). The target edge is therefore approximately four to eight times straighter in this specific local comparison. This describes the three selected local boundaries; the completed comparison above supplies the broader measured context.

Table 6: Same-image comparative morphology from the current report. Texture is variance of a discrete image Laplacian on normalized radiance patches.

Sample	Edge RMS (px)	Relative edge RMS	Texture variance
Target	**0.242**	1.0	8.02×10^{-5}
Nearby clast 1	0.971	4.0	1.64×10^{-4}
Nearby clast 2	1.724	7.1	2.45×10^{-4}
Nearby clast 3	1.953	8.1	2.98×10^{-4}

Transverse radiometry provides an additional test of the exposed body's surface geometry. Replaying the original central-body profile coordinates on the retained radiometric image reproduces a quadratic brightness coefficient of -0.04112 ± 0.00557, with nominal significance $z = 7.38$. Sampling those same five cuts at approximately native-pixel spacing gives nominal $z = 4.80$, and the pooled curvature remains negative when any one cut is omitted. These checks retain evidence of a curved brightness profile across that region.

The current Step 5 selection uses an apex derived from corrected boundary lines. Its origin moves approximately 13 px, shifting the cuts toward the rear and yielding -0.01285 ± 0.01357, with nominal $z = 0.95$. Thus the two statistics describe different sampled regions. The replay, coordinates, and sensitivity checks are retained in `audit/shading-sampling-review/results.json`. Polynomial-fit errors assume independent residuals, while interpolated image samples share information. Profile placement and covariance affect the significance assigned to curvature. Albedo, scattering, dust, and illumination also affect its interpretation as surface shape.

The smoothness, curved terminal boundaries, and tangency-constrained joint fit are complementary constraints. None should be substituted for the others, and the most informative result is their common geometric compatibility.

9. Geometric Implications and Scope

9.1 Planetary-Science Significance

Archived Mastcam imagery permits quantitative tests of centimeter-scale surface morphology when product geometry, radiometry and boundary identity are retained [3, 4]. Here, three different observed boundary families constrain a shared geometric model: the near-terminal curvature, transferred far-arc compatibility, and upper longitudinal boundary. Their relative placement, common projected axis and constrained tangency are tested together. The scientific contribution is the integration of these measurements with explicit prediction tests for this particular feature.

The validation in section 6.3 measures information beyond an overlay. Front/dorsal-derived size, shape and orientation remain frozen while six rear anchors estimate translation; the remaining eighteen stations test the transferred curve. Guarded rear blocks and joint spatial holdouts test prediction across omitted intervals. Withholding the entire dorsal family separately tests how the terminal arcs constrain an unmeasured tangent. These retrospective coordinate tests reuse one acquisition and its original feature selection. Their distinct conditioning explains why their errors differ.

The geological comparison in section 8 establishes that comparable straight segments occur in ordinary rover scenes. Its window distribution measures that single observable. A direct comparison of the complete configuration would also need near/far curvature compatibility, scale, relative placement and upper-boundary tangency. The partial pipe exposures screened here do not supply those jointly observable families. A quantitative physical explanation must additionally reproduce the classified soil contact and shadow geometry while distinguishing exposed material edges from occlusion and illumination boundaries.

An independently constructed geological or manufactured-object surface can be projected through the recorded camera, supplied with explicit terrain and illumination, and evaluated using the same boundary definitions and heldout stations. Finite extent and visibility must accompany tangent consistency, as section 6.7 demonstrates. A process name alone supplies no such reconstruction. Reporting the measured morphology does not require eliminating every possible formation mechanism.

Constant-section engineered elements possess the relations tested by M_1. The measured configuration therefore exhibits manufacturing-like geometric regularity, with support from frozen-shape transfer and guarded prediction. Manufacture remains a morphologically motivated hypothesis alongside physical alternatives whose full scenes require testing. The available boundary geometry leaves composition, provenance and origin unresolved.

Material-boundary classification, subpixel localization, constrained geometry and guarded prediction are established techniques. Their reproducible integration here supplies a case-specific benchmark that can be applied to other rover images. Retained coordinates, exclusions, camera products and scored holdouts allow future comparative-morphology studies to evaluate a proposed explanation against the same observations.

9.2 Measurement and interpretation limits

The validation is retrospective and uses selected coordinates from one acquisition. Neighboring samples share optical and interpolation information, even with coordinate-level guard bands. The nominal reduced chi-square of 33.59 signals an inadequate error/model description; correlation, underestimated localization errors, and model discrepancy are possible contributors, not separately measured explanations. No calibrated origin odds, coverage-calibrated joint confidence interval, or model ranking follows from those nominal weights. Model-specific Birge rescaling is not a common likelihood or a fitted spatial covariance.

The partial terminal arcs constrain visible boundaries rather than both complete hidden end faces. The shared cylindrical model is a constrained image-plane representation, not a unique three-dimensional reconstruction.

Focus-based range and receiving-plane assumptions are conditional. Pixel-to-micrometer conversions quantify projected residual distances at an assumed range, not independently measured machining tolerance or surface roughness. Photometric curvature and its nominal significance depend on the sampled body region and sampling density.

The comparison study measures photometric features, not a population frequency of the full geometry. Its appearance and straightness rankings cannot be converted into probabilities of natural or manufactured origin. Composition, provenance, and a unique concealed shape remain undetermined. Additional observations could address them, but they are not prerequisites for reporting the measured predictive consistency.

10. Reproducibility and Data Availability

The source images, analysis scripts, and numerical outputs are retained in CambrianMinds/martian-cylinder. The original reconstruction base is commit `90fbf8f89ac2634a9dbd2b27f81fc80d68b9943c` (11 November 2026). The present 13 November revision additionally incorporates the predictive validation, completed comparison, and audit corrections. The old base commit alone does not contain this complete revision.

Listing 1: Reproduction from the repository root; no additional image acquisition is required.

```
python -m pip install -r papers/requirements.txt
python audit/sediment-pipe-comparison/scripts/target_benchmark.py
python papers/run_comparison_checks.py
python papers/build_paper_evidence.py
cd papers
xelatex -interaction=nonstopmode -halt-on-error martian-cylinder-technical-paper.tex
xelatex -interaction=nonstopmode -halt-on-error martian-cylinder-technical-paper.tex
cd ..
python papers/verify_journal_revision.py
```

The original extraction scripts are archived under `independent-reconstruction/scripts/`. Primary outputs include `step3_features.json` (stations), `step8_pebble_and_rear_terminus.json` (front/rear definitions), and `step9_joint_model.json` (joint fits and scale profiles). Predictive folds, fitted parameters, residuals, convergence diagnostics, and input hashes are in `audit/predictive-validation/results.json`. The fixed 28-image comparison corpus and its summaries are under `audit/comparative-study/`. The separate pipe inventory, camera-projected sensitivities and endpoint reevaluation are under `audit/sediment-pipe-comparison/`. Figure product IDs, crop bounds, display transforms and exact M_1 parameters are recorded in `papers/figure-provenance.json`.

The accompanying `papers/paper-evidence-manifest.json` identifies exact SHA-256 hashes of the manuscript, scripts, numerical inputs, protocols, and comparison records used for the added results. Image checksums are also collected in `papers/file-integrity.md`; a typeset list of principal revision hashes remains available in `papers/paper-evidence-hashes.tex`. This manifest identifies the working revision without inventing a new committed snapshot or backdating later tests. The figure/data build checks these records and generates the predictive table values directly from JSON.

11. Conclusion

The near rim, exposed far termination and upper boundary exhibit reproducible consistency under common-section, common-axis and tangency constraints. Frozen-shape transfer predicts rear stations beyond those used for registration, while guarded holdouts quantify sensitivity across omitted intervals. These results establish a quantitatively characterized morphological anomaly and manufacturing-like geometric regularity.

Finite-boundary sensitivity also admits a physically tangent tapered segment. The completed straightness comparison and partial pipe inventory do not determine the frequency of the full measured conjunction. Manufacture remains a hypothesis motivated by the geometry; complete three-dimensional shape, composition, provenance and origin remain unresolved. The retained measurements provide a benchmark for testing physical explanations and additional rover observations.

Acknowledgment and provenance note

The calculations and data products synthesized here were developed in the public *martian-cylinder* research repository, using NASA Mars Science Laboratory data. Repository provenance is attributed explicitly; this paper is a synthesis of the recorded investigation, not a claim of independently reacquiring a second rover observation. No NASA institutional endorsement is implied.

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A. Detailed Measurement Inventory

Table 7: Feature-level source coordinates and elementary fits. All coordinates refer to zero-based DRXX pixels.

Feature	Count	Observed region	Output / relation
Near rim	35	x : 1051.52–1083.73; y : 967.15–994.17	Unconstrained ellipse $a = 19.717$, $b = 16.424$, 0.164 px RMS
Far cap	24	x : 1027.91–1039.23; y : 945.40–949.86	Equal-size translated near ellipse, 0.186 px RMS
Upper edge	26	x : 1036.84–1069.03; y : 945.40–962.55	ODR line 28.895°; 0.242 px RMS
Shared fit	85	Combined regions	Strict cylinder: 30.315°, 0.379 px RMS

Feature-selection criteria, including exclusion of the neighboring pebble and separation of the rear cap from soil and longitudinal transitions, are detailed in section 4.

B. Projection and Identifiability Details

For an ideal projected circular end plane with normal aligned to the three-dimensional cylinder axis, the end appears elliptical under oblique viewing. In an orthographic approximation its axis ratio satisfies $b/a \approx |\cos\gamma|$, where γ is the angle between the end-plane normal and the line of sight. The repository's unconstrained near-ellipse ratio $b/a = 0.833$ is therefore compatible with $\gamma \approx 33.6°$, subject to perspective and local projection convention. Its orientation is much less stable than its point-to-ellipse residual because only a partial arc is observed.

The projected position of an unobserved ellipse center is similarly uncertain. Over a short terminal arc, changes in scale and center position can compensate, allowing a one-to-one translation to reproduce the observed curve with only small increases in RMS. This is why rear curvature should be reported as a constraint on the *visible arc*, rather than a direct measurement of the entire concealed end plane. When coupled to a longitudinal surface hypothesis, the same feature gains additional predictive meaning: its center and scale can no longer change independently.

For the strict model, local coordinates of a measured upper-edge point $\mathbf{x}_d = (x_d, y_d)$ are

$$u_d = (x_d - x_f)\cos\theta + (y_d - y_f)\sin\theta, \tag{20}$$

$$v_d = -(x_d - x_f)\sin\theta + (y_d - y_f)\cos\theta. \tag{21}$$

The tangent residual $e_d = v_d + a$ follows immediately from eq. (16). Its sign and the image coordinate orientation are not cosmetic: reversing the orientation of the extracted intensity boundary changes which physical contour is being compared with this equation.